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Spatial Analysis of Common Socio-Economic Factors and Health Outcomes in Chicago Community Areas

Introduction

In this project, I aim to explore how socioeconomic factors like poverty and unemployment impact health outcomes in Chicago, focusing on diabetes-related deaths and infant mortality rates from 2005 to 2011. Living in areas with high poverty and unemployment might limit access to good healthcare, which I think could lead to higher rates of health issues. My hypothesis is that communities with higher rates of poverty and unemployment will have more deaths related to diabetes and higher infant mortality rates. By analyzing this data, I hope to better understand the relationship between economic conditions and health, which could be important for improving public health strategies in similar urban settings.

Methods

The analysis was conducted using the "Chicago Health Indicators 2005-2011" dataset, sourced from the GeoDa website. This dataset includes various health and socio-economic indicators for community areas in Chicago, such as diabetes-related deaths (Diabe_lated), infant mortality rates (Infan_Rate), unemployment rates (Unemp_ment), and percentages of households below the poverty level (Below_level). Data cleaning involved checking for missing or inconsistent entries and addressing them through imputation or removal. No significant data transformations were required. The creation of a weights matrix was used to calculate Moran’s I statistics. This was a rook contiguity weight.

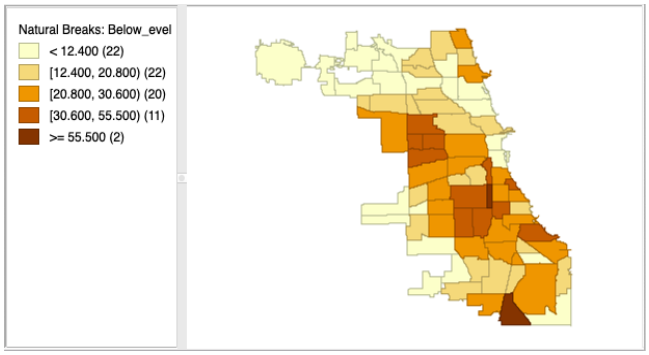
Variables like Diabe_lated and Infan_Rate were used as provided because they are already calculated as rates per population, suitable for spatial analysis. The data covers the period from 2005 to 2011, providing a robust temporal framework for assessing trends over time, although this analysis focuses on the spatial distribution across Chicago's community areas without a specific temporal comparison. Spatially, the data is resolved at the community area level, allowing for detailed geographical analysis within the urban context.

Visual analytics techniques included choropleth maps using Natural Breaks classification to highlight spatial distribution of Diabe_lated and Infan_Rate. Summary statistics from box plots provided a baseline understanding of central tendencies and variability within key variables. A scatter plot matrix was also used to initially asses bivariate correlation. I also ran multiple regression analysis for Diabe_lated and Infan_Rate against all socio-economic variables (unemployment, below poverty level, per capita income, and no high school diploma).

I used a bubble chart to illustrate the relationship between unemployment, infant mortality, and diabetes-related deaths, adding the bellow poverty level statistic for bubble color variation. Moran's I statistic was calculated to assess spatial autocorrelation, indicating potential clustering of similar values within the dataset. A scatter plot matrix was also used to initially asses bivariate correlation.

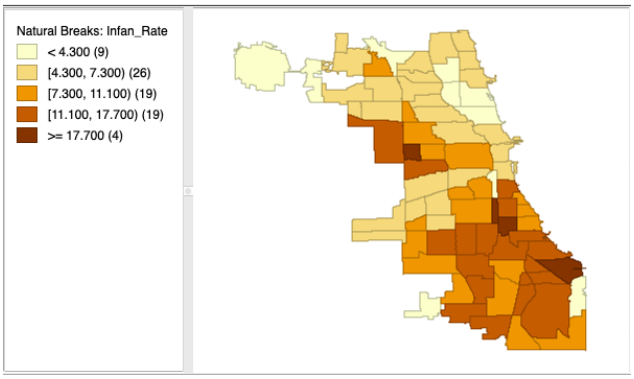
Results

Choropleth maps revealed moderately similar distributions of community areas with high rates of diabetes related deaths and infant mortality, as well as high rates of households below poverty level and unemployment generally for southern community areas. The Northern community areas seemed to have lower rates, specifically near the lakefront.



Percentage of households below poverty level by Chicago Community Area (2005-2011)

Scatter plots showed strong, positive correlation between all variables used. For unemployment vs. infant mortality, a positive relationship is observed, indicating that higher unemployment rates tend to correspond with higher infant mortality. There was a significant correlation with a $p = 0.003$.



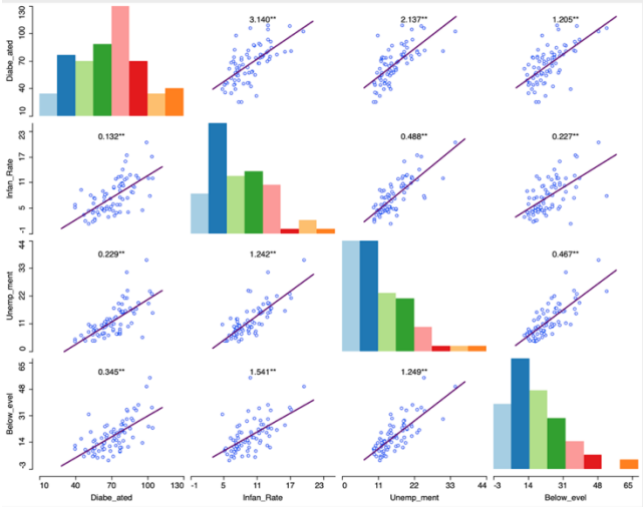
Infant Mortality Rate per 1,000 live births by Chicago Community Area (2005-2011)

For below poverty level vs diabetes related deaths, a similarly positive and significant relationship was observed. However, greater variation among the data points, suggests other factors might also influence diabetes-related deaths.

For Moran's I, infant mortality rate, $I = 0.482$, suggesting moderate spatial autocorrelation, which means that similar values of infant mortality are clustered together geographically. For diabetes related deaths, $I = 0.422$, indicating that areas with high and low diabetes-related deaths are also somewhat clustered

Regression analysis of diabetes-related deaths shows a moderate fit (R^2=0.592), indicating that while socio-economic factors like unemployment and poverty levels are important, they explain about 59% of the variability in diabetes deaths.

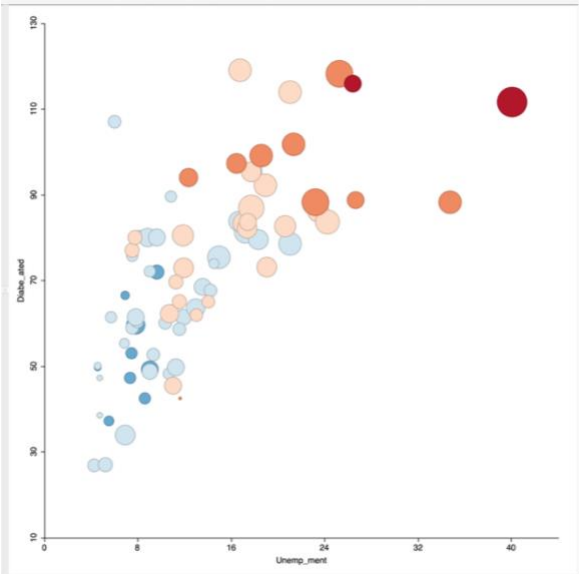
The model for infant mortality rates fits better (R^2=0.641), with unemployment significantly affecting infant mortality ($p = 0.001$).



Scatter plot matrix of selected socio-economic indicators and health outcomes of Chicago community areas (2005-2011)

Results cont.

The final figure, a bubble map shows a relatively normal distribution, with diabetes related deaths increasing with unemployment rates. Small (low infant mortality rate), blue (low percent of households below poverty level) bubbles congregate in the lower corner with few exceptions. The upper right corner of the map has larger, red bubbles.



Std-dev bubble map with unemployment rate (x), diabetes related deaths (y), infant mortality rate (size), and households below the poverty line (color) of Chicago community areas (2005-2011)

Discussion

The analysis reveals clear spatial patterns in health outcomes related to socio-economic status across Chicago. Consistent with the initial hypothesis, areas with higher unemployment and poverty levels demonstrated higher rates of infant mortality and diabetes-related deaths. This finding underscores the influence of socio-economic factors on health, suggesting that lower economic status contributes to poorer health outcomes, potentially due to limited access to healthcare services and higher stress levels associated with financial instability.

The regression analyses provided deeper insights, such as for infant mortality, the stronger predictive power of the model compared to the model for diabetes-related deaths could indicate more direct impacts of socio-economic conditions on infant health. This might be due to factors like prenatal care quality and nutritional access during pregnancy, which are directly influenced by economic conditions. The spatial analysis using Moran's I indicated moderate spatial autocorrelation for both health indicators, confirming spatial dependence. This suggests that similar health outcomes are clustered geographically. Such spatial dependence is critical in planning for health interventions, as it highlights the need to target areas identified as high-risk. However, the presence of spatial heterogeneity, especially in the diabetes data, points to varying impacts of similar socio-economic conditions across different areas, suggesting that local context significantly shapes health outcomes.

The goals were successful, providing a solid analysis of the relationship between socio-economic status and health outcomes in the spatial context of Chicago. Future research could expand on this by incorporating more granular data on healthcare access and environmental factors, or by exploring temporal changes to assess trends over time. One limitation of the current approach was the reliance on broad community area data, which might hide nuanced variation within a community.

Conclusion

This analysis was nothing groundbreaking, but the methods used accurately reflected the hypothesis that the socio-economic condition of Chicago neighborhoods correlates to health outcomes in that area. It also showed that these areas tend to be near each other, with similar outcomes in northern community areas compared to southern ones. I found multiple methods of univariate, bivariate, and multivariate analysis to support my hypothesis, however, finer data granularity may be needed to reveal meaningful spatial patterns. Accessibility to health care, green space, and other aspects of a healthy built environment may also aid in future studies.